

Chemical Engineering | Semester II

# 2071 — Fundamentals of Engineering Chemistry

Credits: 4.5

Contact Hours: 90

Curriculum: REV2026

Comprehensive study of Electrochemistry, Surface Chemistry, and Organic Chemistry fundamentals for Chemical Engineers.

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## Official Source Declaration

This lesson is constructed based on the official **SITTTR Kerala Revision 2026 Syllabus** for Course Code **2071 (Fundamentals of Engineering Chemistry)**. The content covers all modules, subtopics, and practical requirements as specified in the official document.

**Note on Contact Hours:** The syllabus specifies 90 total contact hours. The detailed subtopic breakdown provides approximately 45 Lecture hours and 39 Practical hours (Total 84). This lesson adheres to the detailed topic distribution provided in the official syllabus.

## Course Dashboard

| Attribute       | Details                                   |
|-----------------|-------------------------------------------|
| Program         | Chemical Engineering                      |
| Course Type     | Practicum (Theory + Practical)            |
| Teaching Scheme | 3:0:3:0 (L:T:P:R)                         |
| CIE Marks       | 40 (20 Theory + 20 Practical)             |
| ESE Marks       | 100 (60 Theory + 40 Practical)            |
| Pre-requisite   | 2003B - Chemistry for Technical Practices |

## How to Study This Lesson

- **Theoretical Foundation:** Start with the lecture topics in each module to understand the principles.
- **Practical Integration:** This is a Practicum course. Link the experiments (e.g., Daniel Cell construction) directly to the theory (Electrochemistry).
- **Visual Learning:** Use the SVG diagrams to memorize cell structures and phase diagrams.
- **Malayalam Support:** Refer to the `m1-note` blocks if you find technical English concepts difficult to grasp initially.

## Course Outcomes (CO)

| CO  | Description                                                              | Bloom's Level |
|-----|--------------------------------------------------------------------------|---------------|
| CO1 | Apply electrochemistry and corrosion knowledge to industrial challenges. | Apply         |

| CO  | Description                                                           | Bloom's Level |
|-----|-----------------------------------------------------------------------|---------------|
| CO2 | Apply adsorption mechanisms, colloids, and heterogeneous systems.     | Apply         |
| CO3 | Apply fundamental organic chemistry to solve elementary problems.     | Apply         |
| CO4 | Explore industrially important organic compounds and their reactions. | Apply         |

## Syllabus Coverage

| Module | Title                                               | L Hours | P Hours |
|--------|-----------------------------------------------------|---------|---------|
| 1      | Electrochemistry, Corrosion & Battery Technologies  | 11      | 9       |
| 2      | Surface Chemistry, Colloids & Heterogeneous Systems | 11      | 9       |
| 3      | Chemical Bonding, Carbon & Purification Methods     | 11      | 9       |
| 4      | Important Organic Compounds & Reactions             | 12      | 12      |

## Learning Roadmap

**Step 1:** Electron Transfer (Electrochemistry) & Material Degradation (Corrosion).

**Step 2:** Interface Phenomena (Adsorption) & Complex Mixtures (Colloids).

**Step 3:** Molecular Architecture (Bonding, Hybridization) & Separation Science.

**Step 4:** Synthetic Pathways for Industrial Chemicals (Hydrocarbons, Alcohols, Phenols).

## Module I

# Electrochemistry, Corrosion and Emerging Battery Technologies

Lecture: 11 Hours   Practical: 9 Hours

## 1.1 Electrochemistry Fundamentals

Electrochemistry deals with the relationship between electrical energy and chemical change.

- **Oxidation:** Loss of electrons.
- **Reduction:** Gain of electrons.
- **Redox Reaction:** Simultaneous oxidation and reduction.

**Malayalam support:** Oxidation ഇലക്ട്രോണുകൾ നഷ്ടപ്പെടുന്ന പ്രക്രിയയാണ്. Reduction ഇലക്ട്രോണുകൾ നേടുന്ന പ്രക്രിയയാണ്. ഈ രണ്ടു പ്രക്രിയകൾ ഒരുമിച്ച് സംഭവിക്കുന്ന പ്രതികരണം Redox reaction എന്നറിയപ്പെടുന്നു.

## 1.2 Conductors and Electrolytes

| Type                    | Mechanism         | Example                                       |
|-------------------------|-------------------|-----------------------------------------------|
| Metallic Conductors     | Flow of electrons | Copper, Aluminium                             |
| Electrolytic Conductors | Movement of ions  | NaCl solution, H <sub>2</sub> SO <sub>4</sub> |

**Strong Electrolytes:** Completely ionize (HCl, NaOH).

**Weak Electrolytes:** Partially ionize (CH<sub>3</sub>COOH, NH<sub>4</sub>OH).

## 1.3 Faraday's Laws of Electrolysis

First Law:  $m = ZIt$

Second Law:  $m_1/m_2 = E_1/E_2$

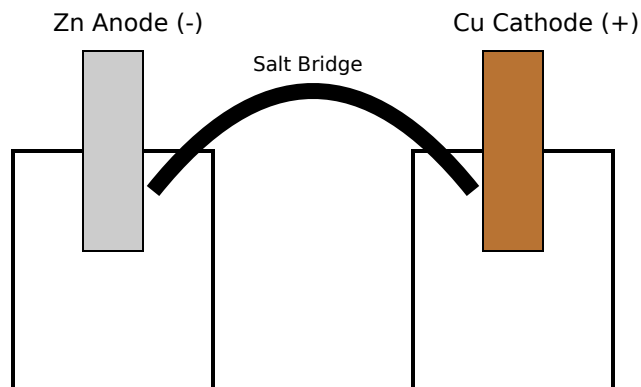
**Given:** Current ( $I$ ) = 2A, Time ( $t$ ) = 10 mins (600s),  $Z$  for Copper = 0.000329 g/C.

**Required:** Mass of copper deposited ( $m$ ).

**Calculation:**  $m = Z \times I \times t = 0.000329 \times 2 \times 600 = 0.3948 \text{ g}$ .

**Answer:** 0.3948 grams.

## 1.4 Electrochemical Cells (Daniel Cell)



*Simplified Daniel Cell Diagram*

## 1.5 Corrosion and Prevention

Corrosion is the gradual destruction of metals by chemical/electrochemical reaction with the environment.

- **Barrier Protection:** Anodic/Cathodic coatings, Anodizing.

- **Cathodic Protection:** Sacrificial Anode method (using Zn or Mg to protect Fe).

## 1.6 Emerging Battery Technologies

- **Li-ion:** High energy density, used in phones/EVs.
- **Na-ion:** Cheaper alternative to Li-ion using abundant Sodium.
- **Solid State Battery:** Uses solid electrolyte instead of liquid; safer and higher capacity.

## Module II

# Surface Chemistry, Colloids and Heterogeneous Systems

Lecture: 11 Hours   Practical: 9 Hours

## 2.1 Adsorption

Adsorption is a surface phenomenon where molecules accumulate on the surface of a solid.

| Feature       | Physisorption      | Chemisorption         |
|---------------|--------------------|-----------------------|
| Force         | Weak Van der Waals | Strong Chemical Bonds |
| Specificity   | Non-specific       | Highly specific       |
| Reversibility | Reversible         | Irreversible          |

## 2.2 Colloids

Particles sized between 1nm and 1000nm.

- **Tyndall Effect:** Scattering of light by colloidal particles.
- **Brownian Movement:** Random zig-zag motion of particles.
- **Electrophoresis:** Movement of particles under an electric field.

**Malayalam support:** Colloids ഏകദേശം 1-1000nm വലിപ്പമുള്ള കണികകളാണ്. ഈ കണികകൾ പ്രകാശത്തെ തിരിയ്ക്കുന്നതിനെയാണ് Tyndall Effect എന്ന് വിളിക്കുന്നത്.



## 2.3 Heterogeneous Systems (Phase Rule)

**Gibb's Phase Rule:**  $F = C - P + 2$

For a Water System (One component,  $C=1$ ):

- Triple Point:  $P=3$ ,  $F = 1 - 3 + 2 = 0$  (Invariant).
- Curves:  $P=2$ ,  $F = 1 - 2 + 2 = 1$  (Univariant).
- Areas:  $P=1$ ,  $F = 1 - 1 + 2 = 2$  (Bivariant).

## Module III

# Chemical Bonding, Carbon and Purification Methods

Lecture: 11 Hours   Practical: 9 Hours

## 3.1 Chemical Bonding

- **Ionic:** Complete transfer of electrons (e.g., NaCl).
- **Covalent:** Sharing of electrons (e.g., HF).
- **Coordinate:** One atom provides both electrons for sharing (e.g.,  $\text{NH}_4^+$ ).

## 3.2 Unique Nature of Carbon

- **Tetravalency:** Carbon has 4 valence electrons.
- **Catenation:** Ability to form long chains or rings with other carbon atoms.
- **Allotropes:** Diamond, Graphite, Fullerenes.

## 3.3 Hybridization

| Type          | Geometry        | Example                           |
|---------------|-----------------|-----------------------------------|
| $\text{sp}^3$ | Tetrahedral     | Methane ( $\text{CH}_4$ )         |
| $\text{sp}^2$ | Trigonal Planar | Ethene ( $\text{C}_2\text{H}_4$ ) |
| $\text{sp}$   | Linear          | Ethyne ( $\text{C}_2\text{H}_2$ ) |

## 3.4 Purification of Organic Compounds

- **Sublimation:** Solid to gas directly (e.g., Camphor).
- **Distillation:** Based on boiling point differences.
- **Chromatography:** Based on differential adsorption (TLC, Column).

## Module IV

## Preparation and Chemical Reactions of Organic Compounds

Lecture: 12 Hours   Practical: 12 Hours

### 4.1 Hydrocarbons

- **Methane:** Decarboxylation of Sodium Acetate with Sodalime.
- **Ethane:** Wurtz reaction (Chloromethane + Na in dry ether).
- **Ethyne:** Calcium carbide + Water.

### 4.2 Important Reactions

**Markovnikov's Rule:** In the addition of HBr to an unsymmetrical alkene, the H attaches to the carbon with more hydrogen atoms.

**Friedel-Crafts Reaction:** Alkylation or Acylation of Benzene using  $\text{AlCl}_3$  catalyst.

**Malayalam support:** Markovnikov's Rule  $\square\square\square\square\square\square$ ,  $\square\square\square$  unsymmetrical alkene- $\square\square\square\square\square\square$  HBr  $\square\square\square\square\square\square\square\square\square\square$ ,  $\square\square\square\square\square\square$   $\square\square\square\square\square\square$   $\square\square\square$   $\square\square\square\square\square\square\square\square\square\square$  H  $\square\square\square$   $\square\square\square\square\square\square\square\square$ .

### 4.3 Alcohols, Phenols and Acids

- **Ethanol:** Prepared by fermentation of molasses. Used as fuel and solvent.
- **Phenol:** Prepared by Dow's process. Undergoes Nitration and Kolbe reaction (to form Salicylic acid).

- **Ethanoic Acid:** Prepared from CO<sub>2</sub> using Grignard reagent. Undergoes HVZ reaction.

## Suggested Student Activities

### Module I & II

- Report on Hydrogen Economy.
- Assignment on Battery recycling.
- Video on Adsorption in water purification.
- Demonstrate Tyndall effect at home.

### Module III & IV

- Assignment on purification methods.
- Model building for Hybridization.
- Identify household organic chemicals.
- Analyze constituents of sanitizers/perfumes.

## Formula Bank

Faraday's 1st Law

$$m = ZIt$$

Gibb's Phase Rule

$$F = C - P + 2$$

Freundlich Isotherm

$$x/m = kP^{1/n}$$

## Visual Library

### Daniel Cell

Essential for understanding electrochemical potential and salt bridge function.

### Water Phase Diagram

Shows the relationship between Ice, Water, and Vapor at different T and P.

## Assessment Methodology

| Component       | Weightage | Details                                |
|-----------------|-----------|----------------------------------------|
| CIE (Theory)    | 20 Marks  | Series tests, Attendance, Assignments. |
| CIE (Practical) | 20 Marks  | Lab performance, Record, Viva.         |
| ESE (Theory)    | 60 Marks  | 2.5 Hours exam covering all modules.   |
| ESE (Practical) | 40 Marks  | 3 Hours internal lab exam.             |

## Module-wise Questions

► **Module 1: Distinguish between Metallic and Electrolytic conductors.**

► **Module 2: What is the Tyndall Effect?**

► **Module 3: Explain Catenation in Carbon.**

► **Module 4: State Markovnikov's Rule with an example.**

## Practice Model Paper

**Note:** This is a practice paper, not an official examination paper.

### Part A (1 Mark each)

1. Define Oxidation.
2. What is a Triple Point?
3. Name an allotrope of Carbon.
4. Write the formula of Methane.

### Part B (3 Marks each)

5. Explain the sacrificial anode method of corrosion prevention.
6. Differentiate Physisorption and Chemisorption.
7. Explain  $sp^3$  hybridization in Methane.

### Part C (7 Marks each)

8. Describe the construction and working of a Daniel Cell with a neat diagram.
9. Explain the Phase diagram of the Water system using Gibb's Phase Rule.

## Quick Revision

### Key Points:

- Anode = Oxidation; Cathode = Reduction.
- $F = C - P + 2$ .
- Colloids: 1nm to 1000nm.
- Carbon: 4 bonds, long chains.

### Exam Checklist:

- Faraday's Law problems.
- Daniel cell diagram.
- Water phase diagram.
- IUPAC Naming rules.

## Glossary

### Electrolyte

A substance that produces an electrically conducting solution when dissolved in a polar solvent.

### Adsorbate

The substance that gets adsorbed on the surface.

### Hybridization

The concept of mixing atomic orbitals to form new hybrid orbitals.

### Isotherm

A curve showing the relationship between pressure and amount adsorbed at constant temperature.

## References



- Principles of Physical Chemistry - Puri, Sharma & Pathania (Vishal Publishing).
- Organic Chemistry - Bahl & Arun Bahl (S. Chand).
- Textbook of Physical Chemistry - P.L. Soni.
- NPTEL Course: 104106119.